Table 1: Generation with observed variables and known structure: mean and sample standard deviation across three seeds. Each entry shows the mean above its ± standard deviation. Validity and coverage are percentages. Bold marks the best mean within each DGP and metric.

| Method | Weighted Discrete | | | Linear | | | Cubic | | | Smoothstep | | | 2D Bridge | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Valid. ↑ | J-MMD ↓ | Cover. ↑ | Valid. ↑ | J-MMD ↓ | Cover. ↑ | Valid. ↑ | J-MMD ↓ | Cover. ↑ | Valid. ↑ | J-MMD ↓ | Cover. ↑ | Valid. ↑ | J-MMD ↓ | Cover. ↑ |
| *Conditional diffusion (generation direction $Z \to X$)* | | | | | | | | | | | | | | | |
| CD | 13.55 ± 2.94 | 0.1467 ±0.0713 | 40.00 ± 5.06 | 83.22 ± 4.51 | 0.0025 ±0.0019 | 78.52 ± 6.64 | 78.12 ± 8.62 | 0.0062 ±0.0036 | 74.35 ± 3.97 | 85.55 ± 3.85 | 0.0034 ±0.0018 | 70.49 ± 5.44 | 80.65 ± 7.18 | 0.0072 ±0.0038 | 73.08 ± 3.27 |
| Compositional CD | 75.50 ± 1.96 | 0.1065 ±0.0072 | 84.78 ± 3.61 | 38.54 ± 1.76 | 0.0139 ±0.0014 | 31.25 ± 4.37 | 31.74 ± 1.32 | 0.0368 ±0.0061 | 27.67 ± 4.42 | 21.21 ± 3.04 | 0.0497 ±0.0037 | 16.06 ± 5.35 | 20.93 ± 2.47 | 0.1054 ±0.0010 | 20.04 ± 3.64 |
| *Extrapolative generation baselines* | | | | | | | | | | | | | | | |
| Additive mean† | 0.00 ± 0.00 | 0.1174 ±0.0093 | 0.00 ± 0.00 | **100.00** ± 0.00 | 0.0088 ±0.0026 | 0.55 ± 0.96 | 0.00 ± 0.00 | 0.1711 ±0.0024 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.0716 ±0.0104 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.3416 ±0.0106 | 0.00 ± 0.00 |
| Additive + residual† | 0.18 ± 0.06 | 0.2322 ±0.0041 | 0.65 ± 0.39 | 7.81 ± 1.53 | 0.1407 ±0.0076 | 6.17 ± 6.23 | 0.00 ± 0.00 | 0.2037 ±0.0008 | 0.00 ± 0.00 | 1.35 ± 0.50 | 0.1523 ±0.0089 | 0.86 ± 0.69 | 0.18 ± 0.20 | 0.3266 ±0.0053 | 0.10 ± 0.10 |
| Fixed composition + residual† | 0.13 ± 0.03 | 0.2353 ±0.0032 | 0.50 ± 0.20 | 0.03 ± 0.06 | 0.2327 ±0.0060 | 0.00 ± 0.00 | 0.02 ± 0.03 | 0.2014 ±0.0141 | 0.00 ± 0.00 | 0.07 ± 0.07 | 0.2548 ±0.0161 | 0.00 ± 0.00 | 0.00 ± 0.00 | 0.3798 ±0.0096 | 0.00 ± 0.00 |
| Rep4Ex-inspired + residual† | 0.85 ± 0.74 | 0.2714 ±0.0658 | 2.87 ± 2.45 | 3.68 ± 3.83 | 0.2490 ±0.1286 | 3.06 ± 3.89 | 0.23 ± 0.35 | 0.2732 ±0.1037 | 0.02 ± 0.03 | 0.39 ± 0.32 | 0.2621 ±0.1050 | 0.02 ± 0.03 | 0.03 ± 0.06 | 0.3648 ±0.1766 | 0.03 ± 0.06 |
| *Classifier-guided diffusion (likelihood direction $X_t \to Z$)* | | | | | | | | | | | | | | | |
| CG: sparse factorized ($s = 1$) | 46.93 ± 7.34 | 0.1027 ±0.0106 | 74.85 ± 3.03 | 5.40 ± 2.81 | 0.0776 ±0.0026 | 1.84 ± 1.49 | 8.92 ± 2.17 | 0.1280 ±0.0034 | 4.07 ± 2.65 | 3.43 ± 1.08 | 0.1096 ±0.0114 | 0.75 ± 0.20 | 4.07 ± 0.63 | 0.2045 ±0.0129 | 3.52 ± 0.81 |
| CG: dense factorized ($s = 1$) | 55.43 ± 6.43 | 0.1073 ±0.0057 | 75.58 ± 2.85 | 7.67 ± 4.64 | 0.0692 ±0.0026 | 2.86 ± 2.14 | 3.45 ± 0.70 | 0.1266 ±0.0039 | 0.65 ± 0.46 | 2.86 ± 1.90 | 0.1073 ±0.0058 | 0.80 ± 0.33 | 1.69 ± 1.21 | 0.2084 ±0.0197 | 1.79 ± 1.71 |
| CG: dense joint ($s = 1$) | 1.47 ± 1.22 | 0.5295 ±0.0298 | 4.27 ± 3.54 | 15.35 ± 11.25 | 0.0359 ±0.0074 | 6.74 ± 6.44 | 13.22 ± 8.30 | 0.0745 ±0.0064 | 5.49 ± 4.16 | 14.75 ± 11.00 | 0.0526 ±0.0078 | 4.57 ± 4.76 | 6.18 ± 6.07 | 0.1360 ±0.0138 | 4.82 ± 4.66 |
| CG: sparse factorized ($s = 2$) | 76.13 ± 5.91 | 0.0530 ±0.0196 | 61.22 ± 1.04 | 24.82 ± 11.92 | 0.0161 ±0.0007 | 7.42 ± 7.76 | 35.74 ± 8.97 | 0.0183 ±0.0024 | 13.85 ± 10.90 | 19.48 ± 10.72 | 0.0383 ±0.0099 | 4.87 ± 2.69 | 36.04 ± 4.75 | 0.0758 ±0.0052 | 30.99 ± 2.84 |
| CG: dense factorized ($s = 2$) | **82.85** ± 5.68 | 0.0633 ±0.0116 | 56.77 ± 3.14 | 29.61 ± 16.03 | 0.0116 ±0.0045 | 9.83 ± 7.13 | 26.16 ± 11.93 | 0.0194 ±0.0003 | 2.67 ± 1.30 | 17.01 ± 14.70 | 0.0423 ±0.0048 | 1.14 ± 1.15 | 5.70 ± 4.56 | 0.0752 ±0.0110 | 4.64 ± 4.15 |
| CG: dense joint ($s = 2$) | 0.45 ± 0.78 | 0.5508 ±0.0336 | 1.42 ± 2.41 | 18.20 ± 26.55 | 0.0111 ±0.0080 | 9.18 ± 15.52 | 34.49 ± 25.69 | 0.0143 ±0.0037 | 13.77 ± 12.87 | 25.72 ± 20.61 | 0.0241 ±0.0018 | 6.35 ± 6.83 | 14.19 ± 19.63 | 0.0394 ±0.0117 | 9.52 ± 12.63 |
| *SEDGE (ours)* | | | | | | | | | | | | | | | |
| SEDGE: no weighting | 12.00 ± 0.15 | 0.2198 ±0.0133 | 29.95 ± 3.03 | 8.98 ± 1.36 | 0.4006 ±0.0066 | 20.82 ± 5.01 | 8.97 ± 1.34 | 0.3650 ±0.0061 | 17.61 ± 3.34 | 8.92 ± 1.34 | 0.4248 ±0.0069 | 17.12 ± 2.88 | 2.46 ± 0.49 | 0.6294 ±0.0069 | 4.10 ± 0.91 |
| SEDGE-LGD ($s = 0.5$) | 38.07 ± 2.28 | 0.1525 ±0.0028 | 70.00 ± 2.51 | 44.11 ± 5.68 | 0.1343 ±0.0121 | 64.81 ± 6.63 | 31.92 ± 2.40 | 0.2048 ±0.0153 | 49.61 ± 5.06 | 47.90 ± 7.27 | 0.1267 ±0.0251 | 60.01 ± 10.11 | 23.67 ± 2.13 | 0.3420 ±0.0225 | 35.40 ± 1.17 |
| SEDGE-LGD ($s = 1$) | 56.80 ± 4.00 | 0.0996 ±0.0107 | 81.08 ± 5.86 | 78.24 ± 2.88 | 0.0104 ±0.0019 | 79.17 ± 1.27 | 60.17 ± 6.17 | 0.0711 ±0.0199 | 64.49 ± 5.27 | 79.57 ± 4.48 | 0.0147 ±0.0063 | 70.05 ± 4.07 | 66.37 ± 2.01 | 0.1153 ±0.0295 | 66.60 ± 1.44 |
| SEDGE-LGD ($s = 2$) | 39.98 ± 19.14 | 0.0272 ±0.0095 | 57.12 ± 18.38 | 77.07 ± 7.78 | 0.0302 ±0.0064 | 62.42 ± 7.84 | 71.89 ± 4.58 | 0.0302 ±0.0024 | 50.91 ± 3.46 | 79.46 ± 4.99 | 0.0328 ±0.0075 | 55.27 ± 9.85 | 76.25 ± 3.09 | 0.0516 ±0.0229 | 56.77 ± 3.07 |
| SEDGE-LWS | 77.55 ± 3.96 | **0.0225** ±0.0183 | **89.97** ± 1.11 | 91.98 ± 1.33 | **0.0018** ±0.0013 | **87.37** ± 3.93 | **91.26** ± 1.27 | **0.0024** ±0.0021 | **84.29** ± 2.46 | **91.55** ± 1.08 | **0.0030** ±0.0007 | **82.23** ± 3.58 | **87.39** ± 3.02 | **0.0016** ±0.0004 | **78.86** ± 1.11 |

Standard deviations use $n - 1$ in the denominator ($n = 3$), not standard errors. Displayed zeros may reflect rounding. Validity requires both specifications and root support; J-MMD measures joint $(X, Z)$ fidelity; coverage uses the reference-based fixed-radius criterion. †Paper-inspired adaptations, not faithful reproductions. Additive mean is a deterministic control, not a posterior sampler. All three original LGD scales and CG scales 1 and 2 are shown; the expanded LGD sweep is excluded. Weighted Discrete is a relaxed Assumption-3 control (conditional exclusion 2.5%/5%); continuous settings meet the working 1% band-conditional criterion. This results-informed selection does not establish uniform superiority. Boldface does not imply statistical significance.